Cantera vs neural networks | Test IC 04

T_0 = 1317.5 K P_0 = 1.244 atm ϕ_0 = 1.984 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.